

Characterization of the pseudopaline transport pathway *via* CntI and MexB in *Pseudomonas aeruginosa* to develop new antibacterial molecules

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ABSTRACT

Metallostasis plays a major role in living systems. It is particularly relevant for pathogen bacteria at microorganism-host interface where « nutritional immunity » is displayed¹. In metal scarce conditions, a common bacterial strategy consists in the biosynthesis of metallophores, their export into the extracellular medium, and the recovery of a metal metallophore complex through dedicated membrane transporters. In *Pseudomonas aeruginosa*, a gram-negative bacteria strongly involved in antibiotic resistance, a broad spectrum nicotianamine-like metallophore named pseudopaline (Pp) has been shown to play a primary role in zinc uptake in a chelating environment². CntI exporter belongs to the drug/metabolite transporter (DMT) superfamily, which is ubiquitous and includes exporters for a remarkably wide range of substrates³.

In order to understand the molecular basis of Pp transport, the structural mechanism of Pp export by CntI across the inner membrane (IM) will be studied. Here we present our preliminary results on the CntI purification, the interaction study between CntI and MexB, and the development of a method for reconstituting our proteins into nanodiscs.

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